**COLOURS OF LIGHT** **Name:** _____

➔ Light is made up of different colours. A spectrum of colours (like a rainbow).

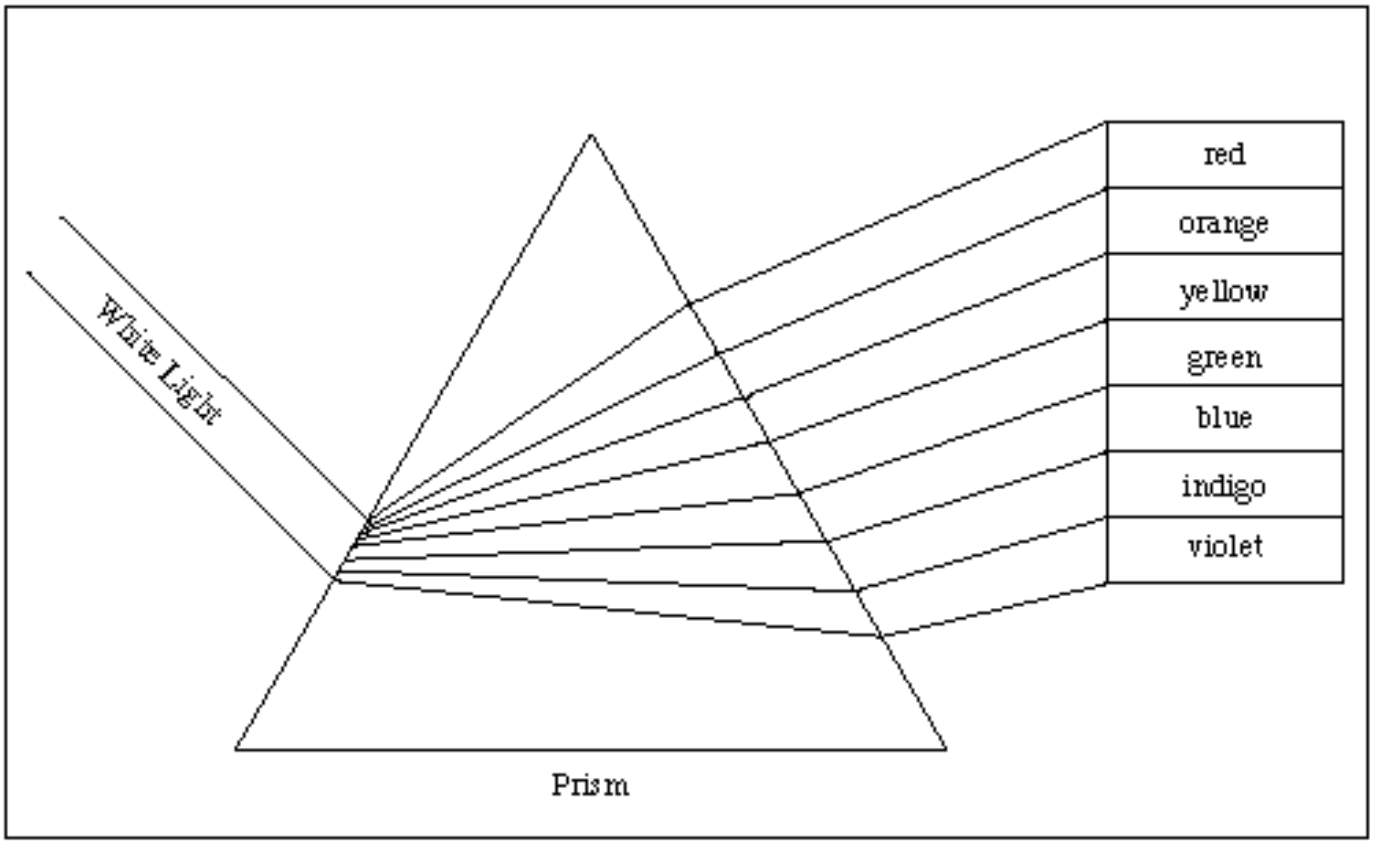
➔ In 1666, Sir Isaac Newton shone a beam of light through a **prism** (a clear plastic 3D triangle) and observed a band of colours appearing on a screen.

➔ The colours were **violet, indigo, blue, green, yellow, orange** and **red**.

➔ From this Newton concluded that **white light** was a mixture of various coloured mixed in certain proportions.

➔ He called this band of colours a **spectrum** and the splitting of white light into colours is called **dispersion**.

***Colour in the spectrum below in the correct colours.***




➔ Each colour of light is different because it has a different wavelength.

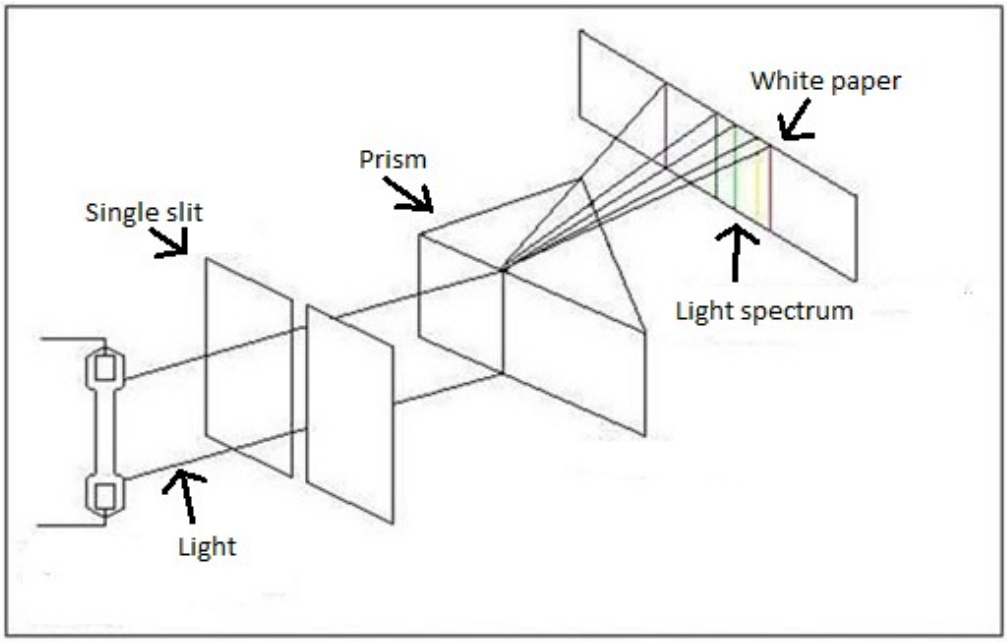
➔ For example, red light has a longer wavelength than violet.

1. White light is made up of **seven** different _____ of light.
2. A **prism** is a special lens that separates white light.
3. If an object reflects yellow light, you see the object as _____.
4. The colours that an object absorbs are the colours that it takes in.
5. A green leaf absorbs all of the colours except for _____.
6. You see an object as the colour of the light it **reflects**.

**Experiment – Creating a spectrum**

Materials – light box, piece of white paper, triangular prism.

1. Arrange the light box so that a single slit of light shines upon a triangular prism.
2. Put the prism on a piece of white paper.
3. Move the prism around until you form a spectrum of light.
3. Trace the prism on the paper and colour in the spectrum.
4. Try using different size slits, different prisms and more than one prism.
5. Each person in the group should draw/colour in a different spectrum.



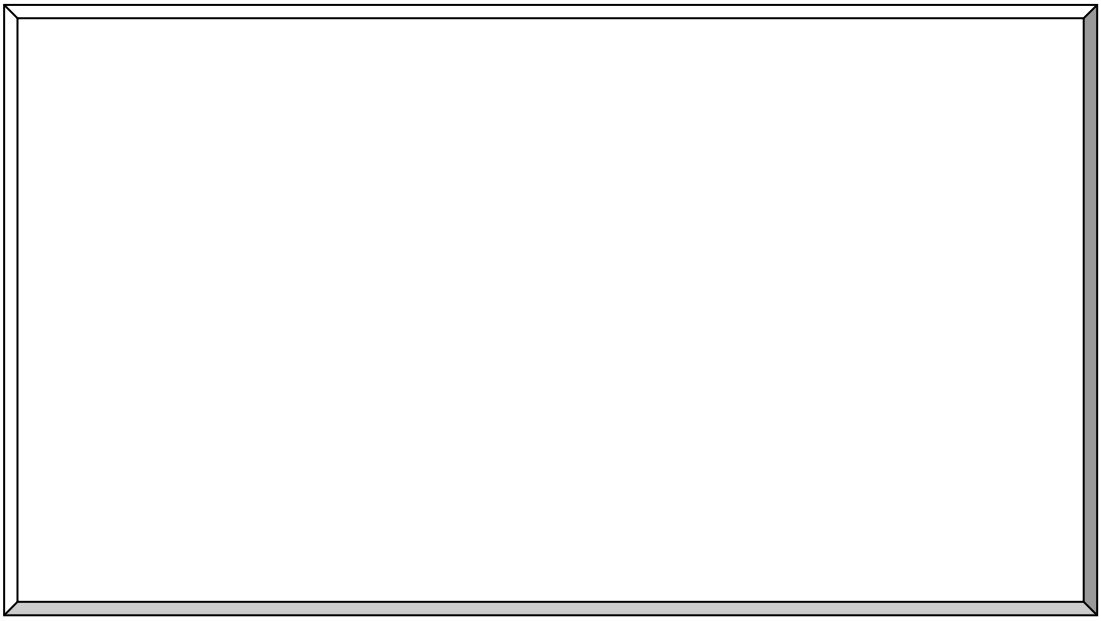

➔ Lights can pass through objects or be reflected by objects.

➔ Light can also go into objects and not come out – this is called being absorbed.

➔ So when light hits something it can:

- Pass **through** the object – for example _____.
- Be **reflected** by the object – for example _____.
- Be **absorbed** by the object – for example _____.

*In the space provided draw a picture of a red apple. Show all the colours of white light hitting the apple like a rainbow. Show that the only colour being reflected is red.*



## TYPES OF OBJECTS

➔ Opaque materials do not allow light to pass through them. They either block or absorb the light.

➔ Transparent materials allow all light to pass through it.

➔ Translucent materials let some light through.

| Material | Opaque, transparent or translucent |
|---|---|
| | |
| | |
| | |
| | |
| | |
| | |
| | |

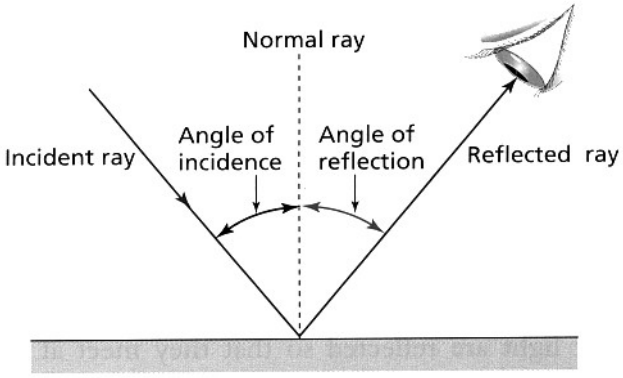
## REFLECTION OF LIGHT

(Fill in the blanks using page 127 of text book)

➔ Reflection is the _____ of light off a _____. Some surfaces appear _____, like a mirror and you can see an _____ in them.
➔ This type of _____ is called _____ reflection.
➔ Regular reflection occurs because the reflecting surface is very _____.
➔ If the surface is not perfectly smooth, then the light will reflect in _____directions and is called _____ reflection.

➔ Objects can be seen because light reflects from them into the eye. The rays of light follow the following rule.

**LAW OF REFLECTION**

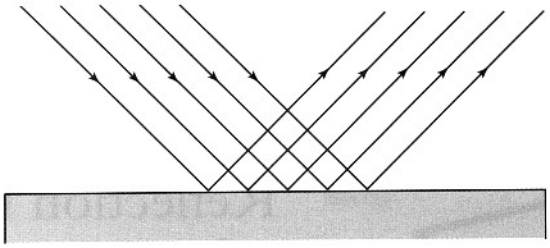



The angle of incidence = the angle of reflection.

**REGULAR AND DIFFUSE REFLECTION**

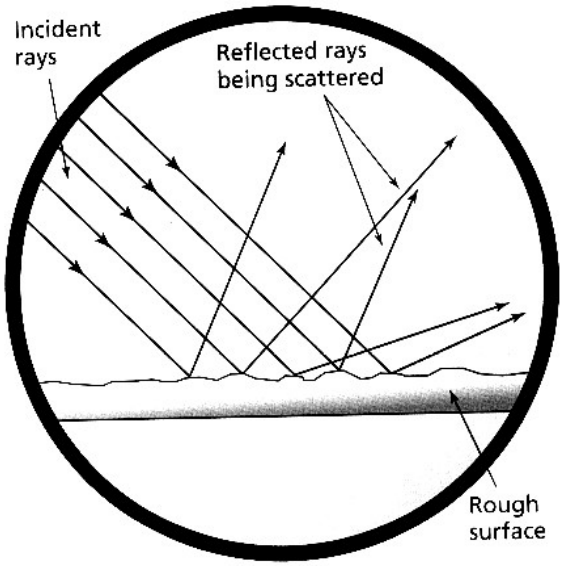
***Regular reflection*** - parallel rays of light that hit a smooth surface are reflected parallel to each other.

e.g. mirror, polished silver



***Diffuse reflection*** - parallel light rays hitting a rough surface are reflected in all directions (the light is scattered).

e.g. newspaper, sandpaper, "matt" paint on walls.



# REFRACTION

➔ Refraction is the bending of light that sometimes happens as light passes through one substance and into another.

Look at a pencil in a beaker of water, see how the pencil looks as though it has been bent?
The light bends as it moves into the water.

**Demonstration – Radiometer**

Materials – radiometer, torch

A simple instrument that is made of four vanes of aluminium foil that spin on a needle point inside a glass globe from which most of the air has been removed. One face of each vane is blackened, while the other side is bright. When placed in sunlight or near any light, the vanes rotate in the direction faced by the bright sides. The black surface absorbs the heat and radiates (gives off) heat while the bright surface reflects heat.

